

ARCHITECTURE CONSTITUTION v1.0

Status: RATIFIED

Mission

Build an AI Manufacturing Intelligence ecosystem through four permanent layers: MiniFlyWire → Noetica → Velith → Mini Prometheus.

Layer 1: MiniFlyWire

Purpose: cognitive research laboratory. Discovers and validates cognitive primitives such as attention, memory, curiosity, reflection, planning, replay, confidence and learning. Output is validated knowledge, not production code. Never imported by other projects.

Layer 2: Noetica

Purpose: Agent Intelligence Platform. Reimplements validated mechanisms into reusable infrastructure: runtime, memory, context, reasoning runtime, planning representation/execution, evaluation, provenance, state substrate, tool interfaces, skills, APIs. Owns mechanisms, never domain knowledge.

Layer 3: Velith

Purpose: Engineering Intelligence. Uses Noetica to solve engineering problems. Owns engineering knowledge, physics world model, CAD, simulation, optimization, engineering verification. Never reimplements platform capabilities.

Layer 4: Mini Prometheus

Purpose: Manufacturing Intelligence. Owns manufacturing planning, factory workflows, robotics, supply chain, MES and production execution. Uses Velith outputs.

Three Permanent Flows

Knowledge Flow: MiniFlyWire → Noetica. Code Flow: Noetica → Velith → Mini Prometheus. Experience Flow: Mini Prometheus → Velith → Noetica → MiniFlyWire.

Constitutional Laws

1. Vision is fixed.
2. One owner per capability.
3. Mechanism != Content.
4. MiniFlyWire never becomes production.
5. Noetica never contains engineering/manufacturing logic.
6. Velith never reimplements Noetica.
7. Promotion requires validation and reimplementation.
8. Platform grows by extraction, not speculation.
9. Strict dependency DAG.
10. Cognition remains core research.
11. Every new feature must be checked against this constitution before implementation.

Decision Protocol

Before any roadmap, module, implementation, refactor, milestone or repository change ask: (1) Which layer owns this? (2) Does it violate ownership? (3) Does it respect the dependency DAG? (4)

Is it mechanism or content? (5) Should it be promoted from research or extracted from Velith? Only proceed if all answers are consistent.

Working Principle (Child Explanation)

MiniFlyWire is the scientist that discovers ideas. Noetica is the factory that turns discoveries into reusable tools. Velith is the engineer that uses those tools to solve engineering problems. Mini Prometheus is the manufacturer that turns engineering solutions into real products. Experience from factories flows back to improve research.